

Erik Henrique Alves Cunha

Full-Stack Software Engineer (Frontend heavy)

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Summary

With 8+ years building production systems with TypeScript, JavaScript, React, Angular, Next.js, Node.js, Python, AWS, SQL, PostgreSQL, MongoDB, REST APIs, CI/CD. Currently shipping LLM, RAG, AI Agents, and GenAI features delivering web applications where predictable behavior, controlled data flow, and operational reliability matter more than novelty. I work at the intersection of frontend architecture and the runtime concerns that decide whether a UI actually holds up: rendering performance, state isolation, error budgets, instrumentation, and the BFF/API contracts that sit just behind the view layer. My strongest impact tends to show up on products with non-trivial data flow, real users on imperfect networks, and teams that need a senior who can de-risk a release without slowing it down.

Skills

Languages: TypeScript, JavaScript (ES6+), SQL
Frontend: Angular, React, Next.js, StencilJS, RxJS, HTML, CSS, SCSS, JSX
State Management: NgRx, Redux, Redux Toolkit
Architecture: Micro-frontends, Web Components, Component-driven architecture, Reactive, Redux, Clean Architecture
Backend: Node.js, Express.js, REST APIs, Webhooks
Databases: PostgreSQL, MongoDB, Microsoft SQL Server, SQLite
Testing: Jest, Playwright, React Testing Library, Karma, Jasmine, MSW, WireMock
Cloud & DevOps: AWS (CodePipeline, CodeBuild, S3, CloudFront, EC2), Docker, NGINX, Jenkins, GitHub Actions
Performance & Accessibility: Core Web Vitals optimization, WCAG, ARIA

Professional Experience

Senior Frontend Software Engineer	Betsson Group	Malta (EU)	03/2025 – Present
- Built frontend reliability and architecture using Angular, TypeScript, RxJS, NgRx, StencilJS, React.js, JSX, Ember.js, Web Components, Micro-frontends, and Nx Monorepo for a PCI-DSS payment and cashier platform processing 40M+ annual transactions and generating €1B+ revenue across 15+ regulated markets for a publicly traded operator - Delivered a shared multi-brand frontend system supporting 8+ languages, 5+ brand variants, and 10+ currencies (including zero-decimal) - Authored a 35+ page frontend knowledge system with architecture diagrams, reducing engineer onboarding time and improving cross-team alignment - Built 12-agent/subagents GitHub Copilot configuration to the team codebase, automating code generation, review, debugging, testing, and architectural validation, increasing engineering throughput and cutting review cycle time across the team - Designed and stabilized stateful multi-step deposit and withdrawal flows in Angular, RxJS, and NgRx, meeting strict audit requirements and regulatory constraints across 15+ jurisdictions - Developed microfrontend using StencilJS, React.js, JSX, enabling shared payment orchestration across multiple production brands simultaneously - Established path-scoped engineering instructions for Copilot (Angular, RxJS, NgRx, accessibility, performance, security) enforced directly through tooling shared across the team - Contributed reusable skills and prompts workflows to the team (component scaffolding, refactoring, Jira automation), improving developer productivity and delivery consistency - Built automated regression protection with Jest and Playwright, isolating APIs via WireMock to safeguard high-risk KYC and payment flows - Implemented automated Copilot session safeguards and lifecycle hooks (Git protections, secret scanning, boundary enforcement) in the team, reducing deployment risk and enforcing engineering best practices			
Senior Software Engineering Consulting	Canon Medical Systems Brazil	Canada - Remote	04/2023 – 02/2025
- Built frontend architecture and delivery using Angular, TypeScript, Nx Monorepo, Clean Architecture (DDD 4-layer), signals-based reactivity, PrimeNG, PrimeFlex, Chart.js, FullCalendar, Bryntum Scheduler, and WebSocket (STOMP/SockJS) for a mission-critical enterprise healthcare operations platform supporting multi-site hospital workflows across a regional network - Reduced JavaScript bundle size by ~33% and CSS bundle size by ~98% through lazy loading, code splitting, and layout refactoring, improving initial load time by 25%			

- Automated infrastructure and releases with Bash scripting, SSH, and NGINX, reducing manual production update time from ~5 minutes to ~45 seconds, rollbacks to ~1 minute, and environment-specific config changes by ~50%
- Improved Time to Interactive (TTI) by ~52% by eliminating waterfall requests via concurrent async loading, combined with OnPush change detection and component-level performance tuning
- Owned the architecture and delivery of the platform end-to-end, applying Clean Architecture (DDD 4-layer) and signals-based reactivity to keep the codebase maintainable as it scaled
- Built data-dense real-time dashboards using PrimeNG, PrimeFlex, Chart.js, FullCalendar, Bryntum Scheduler, and WebSocket (STOMP/SockJS) integrations across 4 complex operational dashboard views
- Achieved WCAG 2.1 AA compliance with Lighthouse accessibility scores near 100 across all production views

React Developer **Grupo SBF (Nike Brazil / Centauro)** **São Paulo, Brazil - Remote** **12/2021 – 04/2023**

- Built frontend performance and reliability using React.js, Next.js, TypeScript, styled-components, Nx Monorepo, SSR/SSG, and Micro-frontends for high-traffic e-commerce platforms (Nike Brazil and Centauro, LATAM's largest sports retailer) serving 8M+ monthly users across 2+ brand properties within a unified commerce platform
- Reduced page load time by ~32% through code splitting, dynamic imports, image optimization, caching strategies, and bundle analysis, improving Core Web Vitals scores by ~12%, primarily driven by LCP improvements
- Improved conversion rate by ~10% by institutionalizing experiment-driven development, delivering and analyzing 20+ A/B experiments with Google Optimize 360 and Google Analytics
- Integrated a WebAR try-on experience into the purchase journey, contributing to measurable uplift in add-to-cart and purchase completion within the feature flow
- Mentored 4 junior engineers in React and testing best practices, establishing review standards and increasing test adoption across the team
- Built and maintained unit and integration test suites using Jest, React Testing Library, and MSW (~70% coverage), protecting 4+ revenue-critical user journeys from regression
- Delivered scalable frontend systems applying SSR/SSG and micro-frontend architecture across 2+ brand properties within a unified commerce platform

Frontend Engineer **Encora Inc. (Daitan Group)** **Campinas, Brazil** **01/2021 – 12/2021**

- Built enterprise frontend features using Angular, TypeScript, RxJS, NgRx, reactive forms, Jest, Karma, Jasmine, CI/CD, AWS CodePipeline, CodeBuild, S3, and CloudFront as part of the engineering team delivering VMware Pathfinder – VMware's official global product test-drive and hands-on labs discovery platform serving 2.1M+ cumulative labs delivered across multi-language, multi-region users worldwide, covering vSphere+, vSAN+, Aria, Tanzu, Horizon Cloud, NSX, and Cloud Foundation product lines
- Built automated testing pipelines using Jest, Karma, and Jasmine, achieving ~70% coverage on core business workflows and significantly lowering regression risk during releases
- Introduced structured client-side state management through NgRx, organizing actions, reducers, selectors, and effects to stabilize async data flows and reduce UI inconsistency across feature modules
- Elevated UI accessibility by implementing semantic markup, ARIA roles, keyboard navigation patterns, and smooth CSS transitions across multiple application views
- Integrated Zoom APIs to support in-app meeting scheduling and embedded collaboration workflows for VMware customer engagement sessions
- Contributed to CI/CD delivery pipelines using AWS CodePipeline and CodeBuild, with static assets deployed via Amazon S3 and CloudFront

Frontend Engineer **Zup Innovation (Itaú bank)** **Campinas, Brazil** **04/2020 – 01/2021**

- Built frontend systems using Angular, TypeScript, RxJS, HTML, SCSS, Web Components, Karma, Jasmine, AWS CodePipeline, CodeBuild, S3, and CloudFront for digital banking platforms at Zup Innovation, the Itaú Unibanco-owned tech arm building secure, scalable distributed systems for Latin America's largest bank – serving 70M+ customers across 18+ countries through digital banking workflows
- Led modular UI architecture initiatives, designing reusable Angular component libraries and Web Components that enabled micro-frontend integration across multiple banking applications
- Introduced automated regression protection using Karma and Jasmine, expanding test coverage across core transactional interfaces and shared components
- Applied accessibility and compliance standards (semantic HTML, ARIA roles, keyboard navigation) across banking workflows requiring regulatory validation
- Integrated Angular clients with banking backend services via REST APIs, ensuring predictable data contracts and reliable client-server communication

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| Frontend Engineer | Venturus | Campinas, Brazil | 11/2019 – 04/2020 |
| <ul style="list-style-type: none"> - Built operational dashboards using Angular, TypeScript, RxJS, HTML, SCSS, Karma, Jasmine, and Jenkins CI/CD for CCR AutoBAn (subsidiary of CCR Group / Motiva, Brazil's largest highway concessionaire) managing the 320 km Anhanguera-Bandeirantes (SP-330 + SP-348) toll and transportation system, one of Brazil's most strategic highway corridors connecting Greater São Paulo to Campinas and the country's interior - Engineered modular components and reactive data flows enabling real-time monitoring, filtering, and operational analysis across multiple dashboard views processing high-frequency telemetry and transactional data streams - Built client-side CSV export and parsing tools, allowing operational teams to generate large reporting datasets directly from dashboard applications - Introduced automated testing practices using Karma and Jasmine, improving regression protection across complex data-driven UI components - Collaborated with DevOps teams supporting Jenkins CI/CD pipelines to validate builds, maintaining release stability | | | |
| Full Stack Engineer (MEAN Stack) | Venturus | Campinas, Brazil | 02/2019 – 11/2019 |
| <ul style="list-style-type: none"> - Led development of a foreign trade advisory platform using Angular, Node.js, Express, PostgreSQL, MongoDB, Docker, Docker Compose, and Sentry for CGO Assessoria, a 30+ year Brazilian import/export and agribusiness customs advisory specialist serving clients across multiple sectors nationwide, supporting compliance-driven workflows for international trade documentation of seeds, fertilizers, agricultural machinery, and other regulated agribusiness inputs - Resolved critical backend performance bottlenecks in reporting and data-intensive endpoints, reducing reporting API execution time by ~97.5% (~40s → <1s) and improving other endpoints from ~2s to ~200ms through query redesign, indexing strategies, and optimized ORM access patterns - Redesigned core data architecture, migrating document-heavy workloads from MongoDB to PostgreSQL with relational modeling and optimized SQL indexing - Developed a Node.js ETL pipeline to migrate production datasets between databases with schema transformation and validation safeguards from Mongo (NoSQL) to PostgreSQL (SQL), normalizing and sanitizing all data - Introduced centralized observability using Sentry, improving error detection and production debugging across backend and frontend services - Standardized development environments using Docker and Docker Compose, improving environment reproducibility for the entire engineering team | | | |
| Mobile Developer | MB Labs | Campinas, Brazil | 09/2018 – 02/2019 |
| <ul style="list-style-type: none"> - Architected and shipped SM Aprendizagem, a cross-platform educational app using Ionic, Angular, TypeScript, Electron, SQLite, and REST APIs for Edições SM, the Brazilian arm of Grupo SM (a Spanish educational publishing group present in nearly 80 countries, founded 1937) — delivering students, families, and teachers offline access to didactic books and digital content across 5 operating systems (Android, iOS, Windows, macOS, Linux) - Reduced platform delivery cost by ~80% versus maintaining separate native codebases by architecting a shared Ionic + Angular codebase achieving ~90% logic reuse across all 5 platforms, consolidating mobile and desktop into a single release pipeline - Cut desktop build time by ~40% versus maintaining independent desktop projects by integrating Electron for Windows, macOS, and Linux, exposing native filesystem and window APIs - Implemented offline user event tracking based on book interactions, recording reading progress, chapter navigation, and engagement events locally via SQLite and syncing automatically with the backend API on reconnection, with zero data loss across sessions - Designed a REST API sync engine with incremental updates and conflict resolution, keeping client-server state consistent across 3+ device types with zero data loss on reconnection | | | |
| Full Stack Developer | MB Labs | Campinas, Brazil | 07/2018 – 09/2018 |
| <ul style="list-style-type: none"> - Architected backend services using Node.js, Express, Firebase Cloud Messaging (FCM), AWS EC2, NGINX, Route 53, S3, and Amazon RDS for the Hondana INDICA enterprise chatbot platform — a Brazilian B2B corporate training / HR-tech product integrated with Facebook Workplace and adopted by enterprise clients including Cia Hering (reporting 450% greater participation and engagement vs. prior training methods) and KPMG Business School - Designed asynchronous webhook-based event pipelines enabling reliable message processing and integration with external systems - Implemented batch push notification delivery with retry logic to ensure reliable message distribution at scale | | | |

- Configured full cloud infrastructure for the chatbot platform: provisioned compute instances, deployed reverse proxy, managed DNS, hosted static assets, and configured relational database connectivity
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| Frontend Engineer (Vue.js) | Monde Sistemas | Americana, Brazil | 10/2017 – 05/2018 |
|-----------------------------------|-----------------------|--------------------------|--------------------------|
- Built responsive frontend features using Vue.js, JavaScript (ES6+), HTML5, CSS3, Stylus, Jest, Sinon, REST APIs, JSON:API specification, Webpack, npm, Git, Git Flow, and Agile/Scrum practices for Monde Sistemas — a Brazilian B2B SaaS travel-agency management (CRM/ERP) platform used by 2,200+ travel agencies across Brazil
 - Developed reusable Vue.js components, single-page application (SPA) views, and shared UI design-system patterns across production web applications, contributing to maintainable frontend architecture and consistent user experience
 - Integrated RESTful APIs following the JSON:API specification, implementing asynchronous client-side data fetching, state handling, and error management for CRM/booking workflows
 - Introduced automated unit testing with Jest and Sinon (mocks, stubs, spies), improving code reliability, regression coverage, and developer confidence across shared frontend modules
 - Collaborated cross-functionally with UX/UI designers, backend engineers, QA, and product within Agile/Scrum sprints and Git Flow branching workflows, delivering production-ready features through code review, pull requests, and continuous integration

Education

Web Development Specialist Co-op Diploma	Cornerstone International Community College of Canada (CICCC)	Vancouver Canada	08/2023 – 02/2024
Bachelor's of Information Systems	Centro Universitário Salesiano de São Paulo (UNISAL)	Campinas Brazil	01/2015 – 06/2020

Certifications

- **Verified International Academic Qualifications** — World Education Services (WES), 2022
- **IELTS General Training** — Band 6.5 (C1 Speaking and Listening), 2023
- **Angular Developer Certification** — Alain Chautard (Google Developer Expert in Angular), 2024

Languages

- **Portuguese** — Native
- **English** — C1
- **French** — A2

Projects & Showcase

- [Nike Stores Finder](#): Location-based web application integrating Google Maps Geolocation and Directions APIs to discover nearby retail stores and calculate routes. Demonstrates independent system design, API integration, and client-side state management

Volunteering

- **DevOpsDays Campinas — Organizer (2024)**
- Led the full speaker sourcing, evaluation, and selection process across DevOps, cloud infrastructure, platform engineering, and software architecture submissions
- Curated a technical program of 10+ talks and lightning sessions, balancing depth and accessibility for a mixed practitioner audience
- Coordinated with speakers, sponsors, and co-organizers across the event cycle, from CFP launch to post-event wrap-up